



Brain Berry

LEARN SMARTER. GROW BRIGHTER.

Theme ID: 2 - Content Creation: Entertainment & Gaming

Team name: **Shades Of Dark**

College : **MS Ramaiah Institute of Technology**

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Problem statement

Today, Autism Spectrum Disorder (ASD) is a major neurobehavioral problem of global concern. As of 2020 CDC reports, 1 in 36 children in USA come under the spectrum and autism occurs across all racial, ethnic and socioeconomic groups. The estimated prevalence in India is 1.12 (0.74-1.68) per 100 children aged 2-9 years i.e., **1 in every 68 children** is affected with autism

Source : [indianpediatrics](https://www.indianpediatrics.org)

Despite the growing prevalence, **standardized teaching methods in India and worldwide frequently fail to meet the unique needs of neurodiverse children**—including those with ASD and other neurodevelopmental disorders (NDDs). Evidence from educational research highlights that these inflexible approaches often lead to low engagement, missed opportunities for developing social-emotional and executive skills, and frustration among both learners and educators.



Our Solution

BrainBerry offers an innovative GenAI-assisted therapeutic game platform purpose-built for neurodiverse children. By combining advanced AI and pedagogical research, BrainBerry bridges the gap in personalized, effective learning for children with Autism Spectrum Disorder (ASD) and other neurodevelopmental disorders.

- A **“Game Mold”** model separates pedagogy (immutable templates) from **personalization** (AI-generated game assets), ensuring consistency while tailoring content to each child’s interests.
- **Conversational 3D avatars** provide text and voice chat with real-time lip-sync, creating warm, motivating interactions during play.
- **Built-in** assignments, sessions, and analytics support supervision, progress tracking, and longitudinal outcomes.



Unique Selling Proposition

- 1 **Evidence-informed templates meet GenAI personalization:** faster content creation for educators without diluting pedagogy.
- 2 **Guard-railed AI:** schema-validated generation keeps experiences safe, age-appropriate, and auditable.
- 3 **Conversational engagement:** avatar voice/text chat + real-time lip-sync increases motivation and session adherence.
- 4 **Democratizing Content Creation:** Our no-code portal empowers non-technical users (therapists, teachers) to create specific and relevant content for their students.
- 5 **Human-AI Collaboration:** The AI generates the content, but the human educator guides the learning goals and interprets the results, creating a powerful synergy.

WHO WE ARE

We game smart. We support smarter.

BrainBerry welcomes all neurodiverse learners. From first-time players to advanced problem-solvers.

Our mission: Build confidence through personalized gaming. Push boundaries. Share success.

Rapid 2-week development for maximum therapeutic impact



SEE BRAINBERRY IN ACTION

Real Progress

Transforms learning through personalized experiences.

Adapts to every learning style

Engages students and therapists

Increases session activities

Improves student engagement

TRY IT NOW



COMMUNITY

REAL STORIES



8 min read

Sensory Processing

Guide to helping your child navigate sensory challenges through gaming.

READ MORE



5 min read

Success Stories

Emma's First Breakthrough

How personalized games helped Emma overcome social anxiety and build confidence.

Jennifer K. (Parent)
Aug 23, 2025

READ MORE



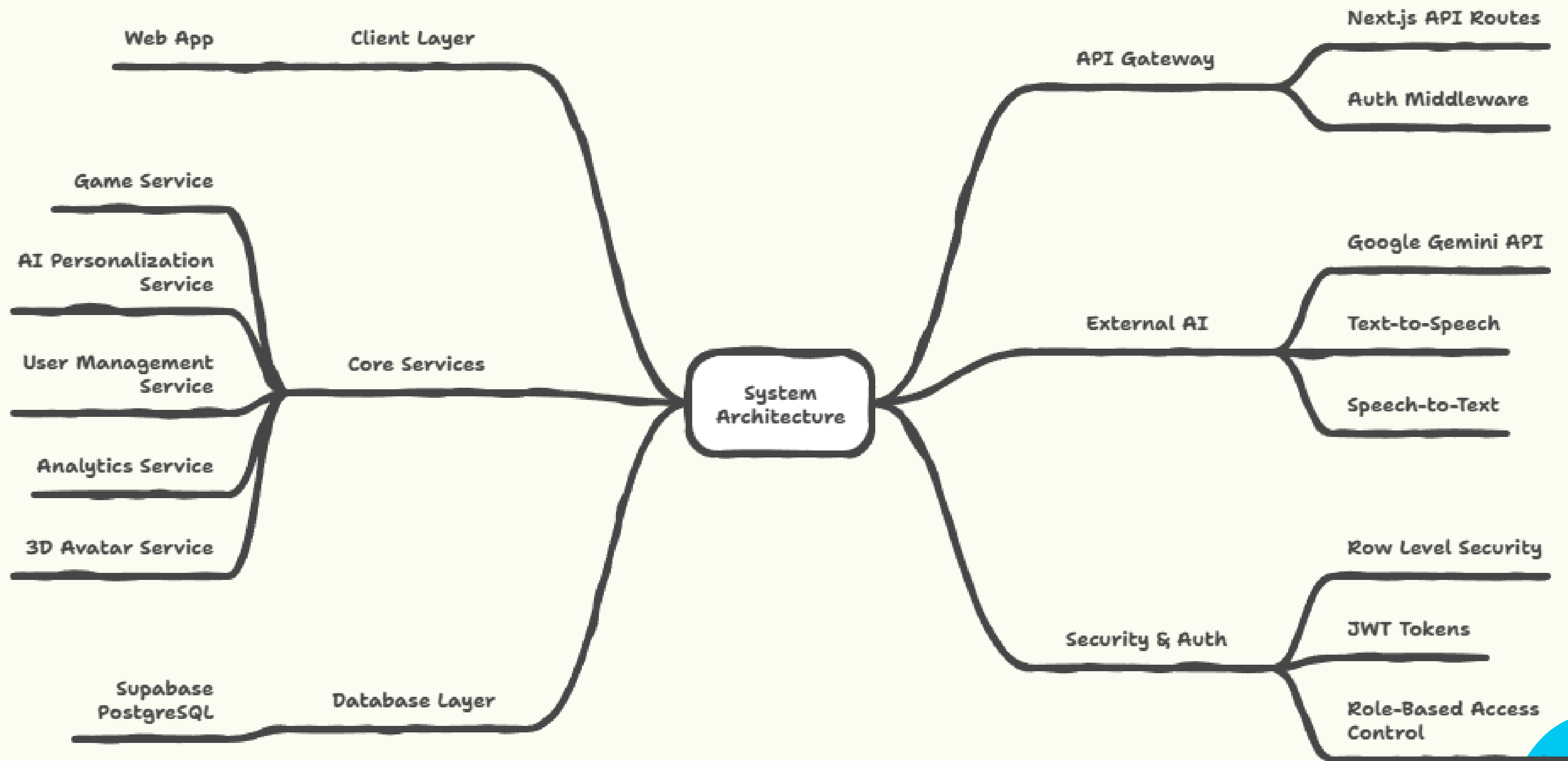
Educational

Building Executive Function

Research-backed strategies for improving working memory and attention through gaming.

BrainBerry Team
Aug 20, 2025

System Architecture

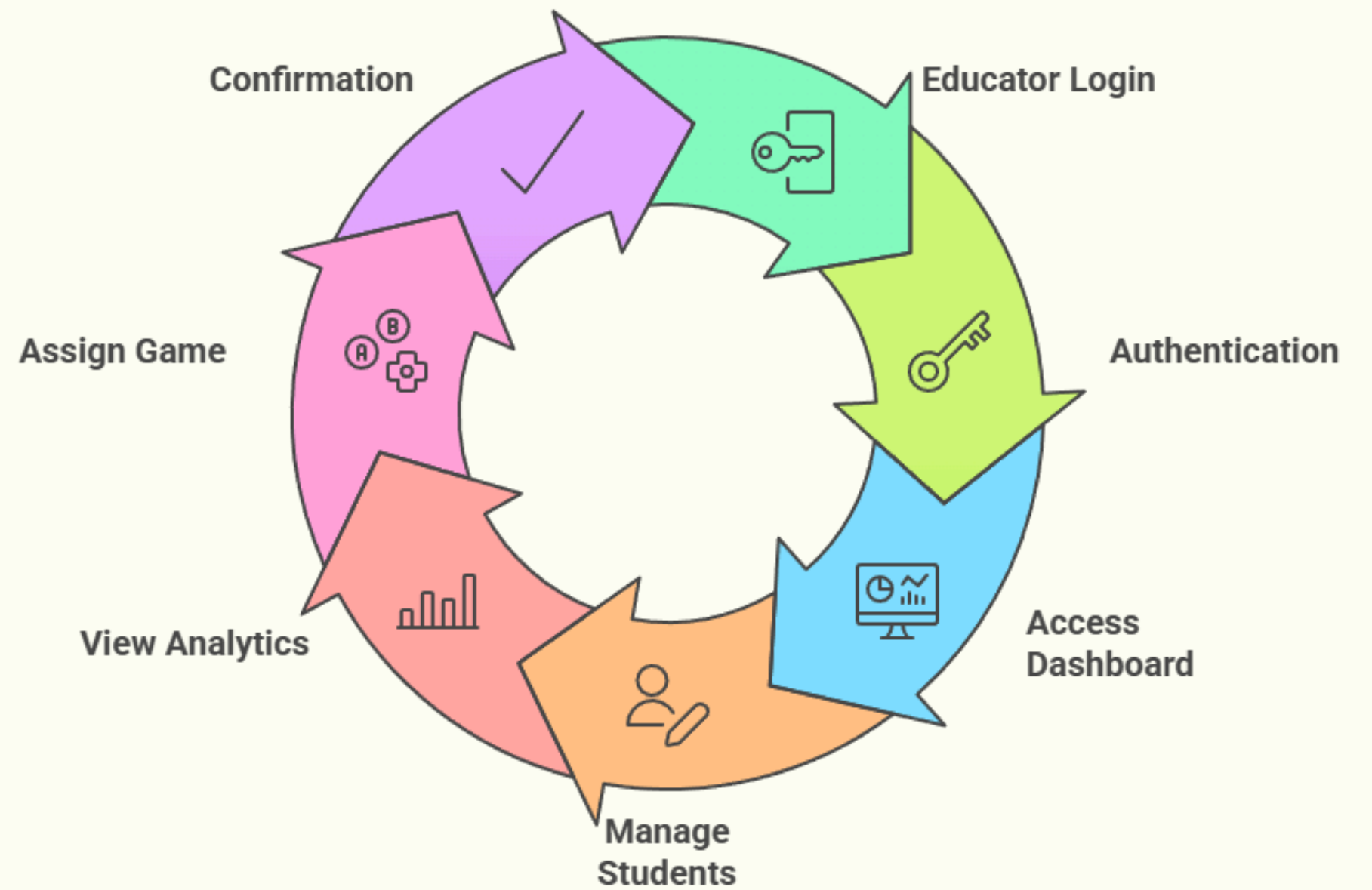


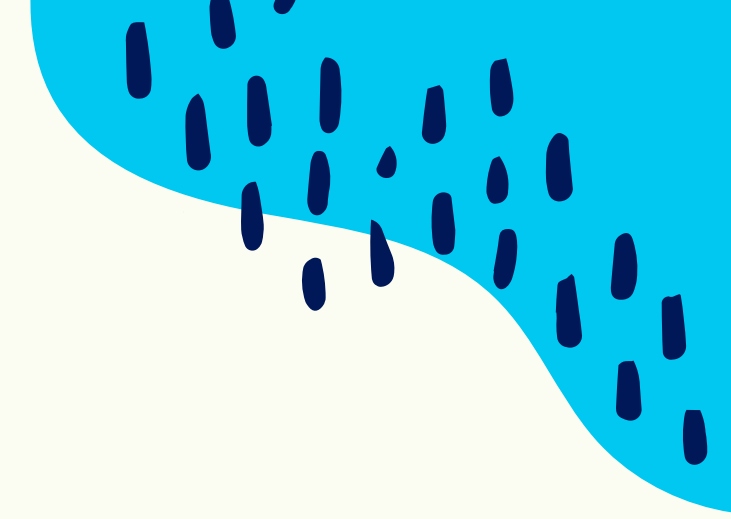
User Flow

Child's Interactive Learning Cycle



Educator Dashboard Cycle





Tools and Tech Stack

Frontend & UX

- Next.js 15 (App Router)
- React 19
- Tailwind CSS
- Lucide Icons

3D Avatar& Realtime Audio

- three.js
- @react-three/fiber
- ReadyPlayerMe avatars
- wawa-lipsync

AI tools

- Google Gemini (TTS, Live/streaming)
- face-api.js



Backend & Data

- Supabase (Postgres, Auth, Storage, RLS)
- SSR client for secure API access

Validation & Safety

- Zod schemas
- structured API responses
- centralized error handling and logging



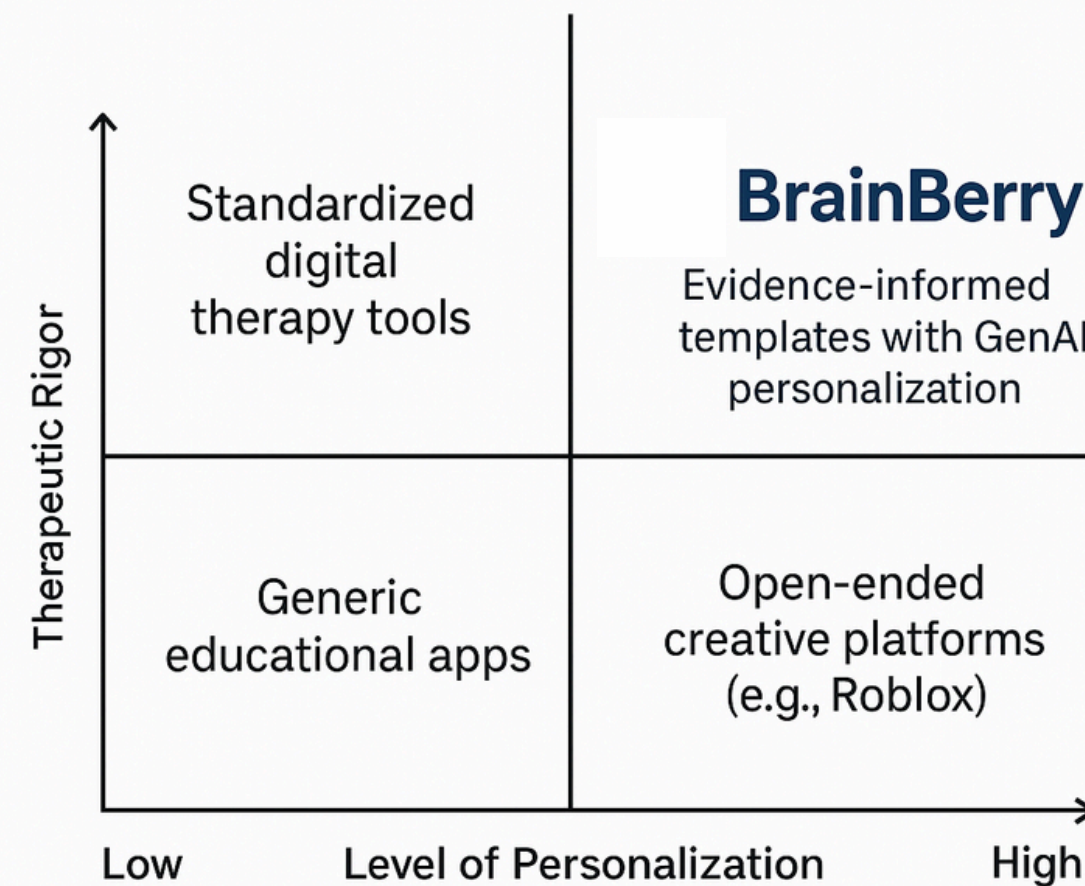
Tooling & QA

- TypeScript
 - ESLint
 - Recharts
- 

Our Market: Users, Customers, and Competition

Our Users & Customers

- **Primary Users:** Neurodiverse children with conditions like ASD and ADHD.
- **Primary Customers:** Therapists, special educators, schools, and clinics who need effective, personalized content.
- **Secondary Customers:** Parents seeking supplementary at-home tools.





Business Plan

Market entry (9-12 months)

- Run paid pilots with 5–10 clinics/schools; prove outcomes (time-on-task, mastery, adherence) and publish 3 case studies along with marketing.

Monetization

- Tiered SaaS per child/seat with add-ons (advanced analytics, premium content packs, voice minutes); volume discounts for districts/networks.

Distribution & partnerships

- Channel deals with therapy networks and special-ed org; content/IP partners for themed molds; explore employer/insurer wellness benefits.

Compliance & scale

- Achieve COPPA/GDPR and SOC2 readiness; multi-region hosting; outcomes reporting for payers; formalize procurement (security + DPIA packs)

Global scale note:

- Global ADHD pooled prevalence $\approx 8\%$ in children/adolescents and global ASD $\sim 1\%$ indicate substantial international expansion opportunity after domestic validation.

India pilot rationale:

- Multi-site studies show site-specific NDD prevalence from $\sim 2.9\%$ up to $\sim 18.7\%$ (depending on age and region); pooled estimates across sites point to $\sim 9\text{--}13\%$ in studied age groups.
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Thank You!

